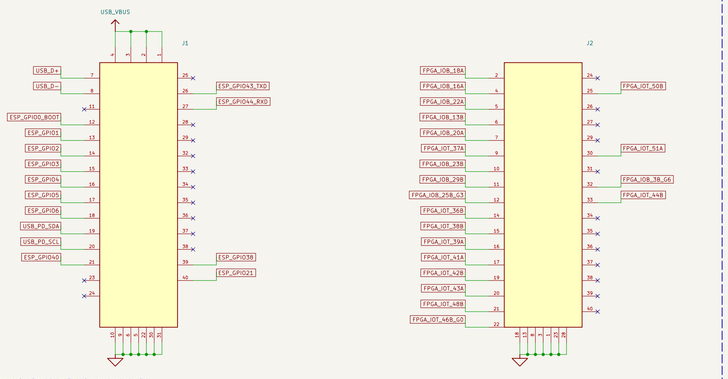
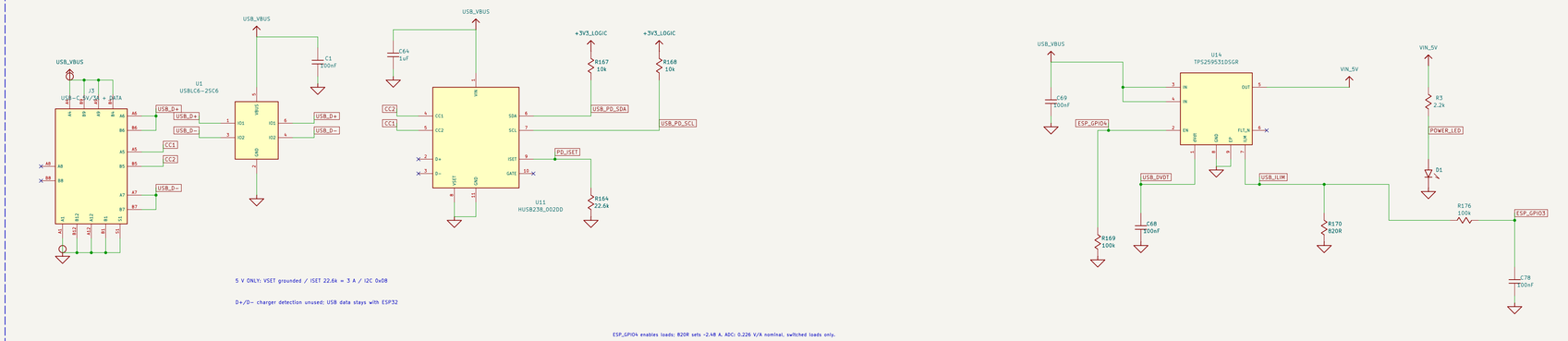
MODULE v1 / exact programmer mating interface

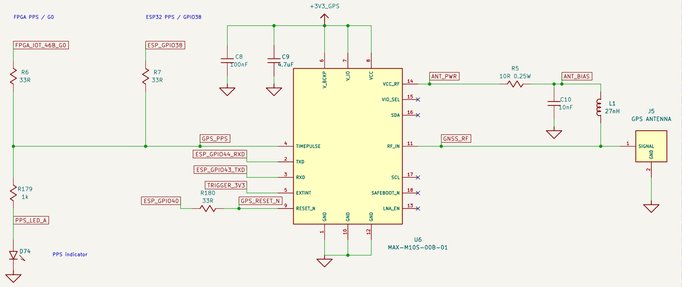


25 mm mating spacing / module top flush / no reset or boot switch

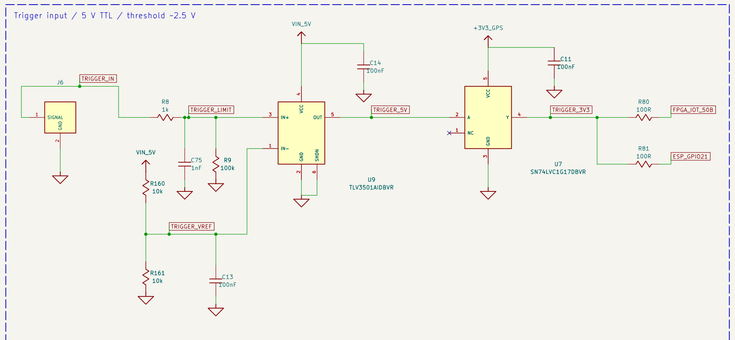
USB-C PD / 5 V, 3 A sink and protected power



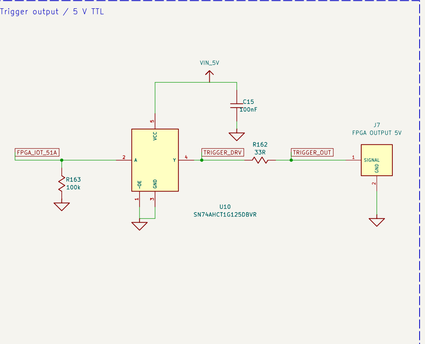
GNSS / PPS reference and active antenna bias



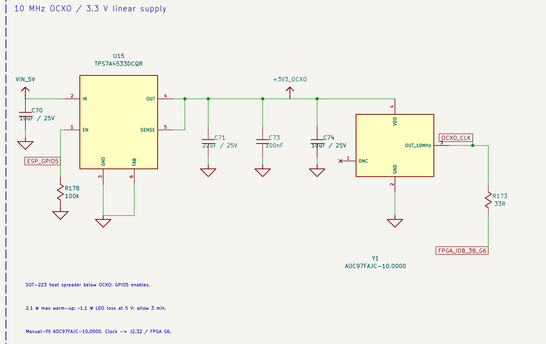
Trigger input / 5 V TTL / threshold ~2.5 V



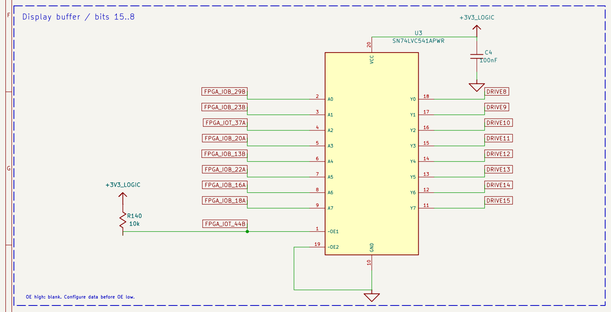
Trigger output / 5 V TTL



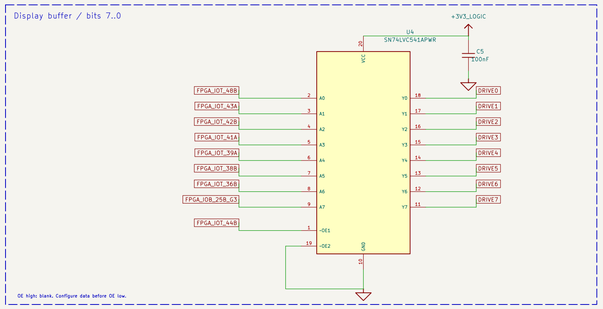
10 MHz OCXO / 3.3 V linear supply



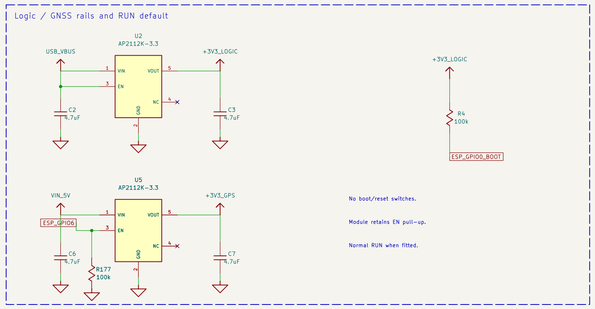
Display buffer / bits 15..8



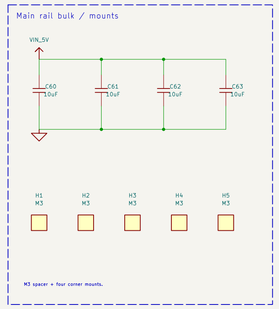
Display buffer / bits 7..0



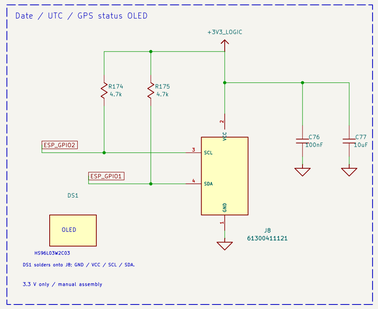
Logic / GNSS rails and RUN default



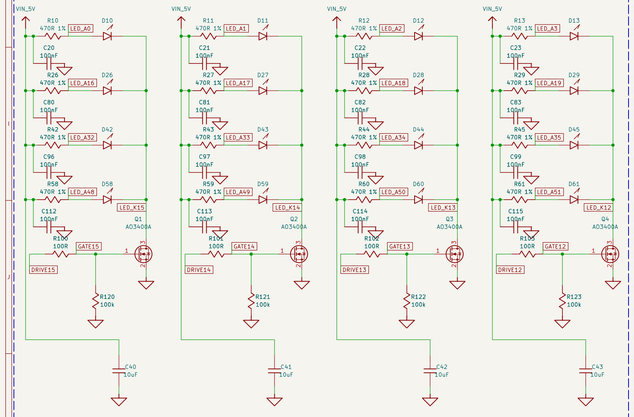
Main rail bulk / mounts



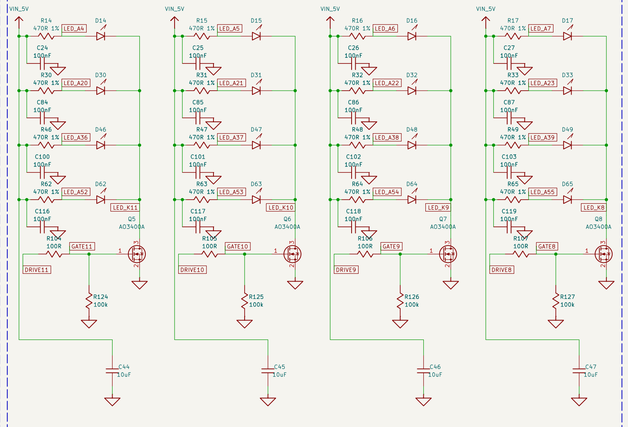
Date / UTC / GPS status OLED



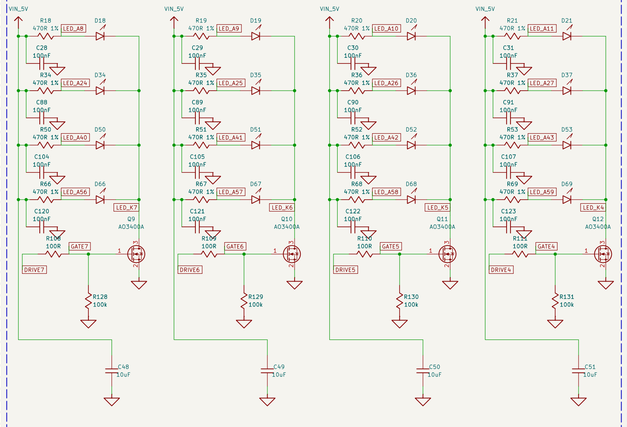
OPTICAL BANK 1 / bits 15..12 / four identical rows



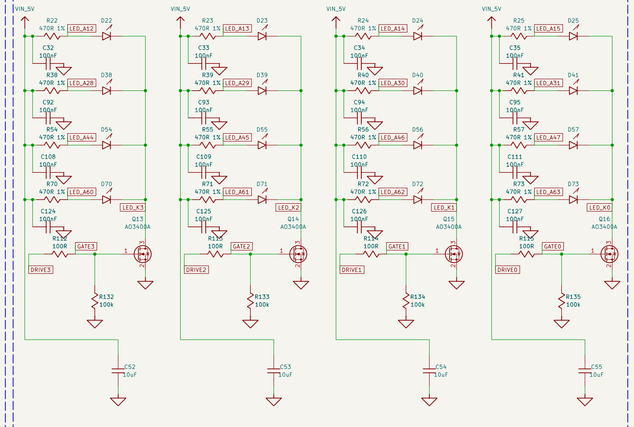
OPTICAL BANK 2 / bits 11..8 / four identical rows



OPTICAL BANK 3 / bits 7..4 / four identical rows



OPTICAL BANK 4 / bits 3..0 / four identical rows



Diagnostic probe pads / functional names on PCB / bare copper, no assembly parts

TP1 // VBUS — USB_VBUS
TP2 // 5V_LOAD — VIN_5V
TP3 // 5V_LOGIC — +5V_LOGIC
TP4 // 3V3_GPS — +3V3_GPS
TP5 // 3V3_OCXO — +3V3_OCXO
TP6 // 3V3_LOGIC — +3V3_LOGIC
TP7 // LOAD_EN — ESP_GPIO6
TP8 // DCXO_EN — ESP_GPIO5
TP9 // GPS_EN — ESP_GPIO4
TP10 // PD_SDA — USB_PD_SDA
TP11 // PD_SCL — USB_PD_SCL
TP12 // I_MON — ESP_GPIO3
TP13 // PPS_SMC — GPS_PPS
TP14 // PPS_FPGA — FPGA_IOT_48B_GO
TP15 // PPS_ESP — ESP_GPIO38
TP16 // CLK_FPGA — FPGA_IOB_38_G6
TP17 // TRG_DVR — TRIGGER_DRV
TP18 // TRG_CMP — TRIGGER_CMP
TP19 // TRG_5V5 — TRIGGER_5V5
TP20 // TRG_3V3 — TRIGGER_3V3
TP21 // OUT_FPGA — FPGA_IOT_51A
TP22 // OUT_DRV — TRIGGER_DRV
TP23 // OUT_CMP — TRIGGER_CMP
TP24 // TRG_FPGA — TRIGGER_FPGA
TP25 // GPS_TX — ESP_GPIO44_RXD
TP26 // GPS_RX — ESP_GPIO43_TXD
TP27 // GPS_RST_N — GPS_RESET_N
TP28 // LED_DE_N — FPGA_IOT_44B
TP29 // OLED_SCL — ESP_GPIO2
TP30 // OLED_SDA — ESP_GPIO1

bp-test
Sheet: /
File: calibrator.kicad_sch
Title: GPS optical time calibrator / module v1 carrier
Size: A1 | Date: | Rev: A draft
KiCad E.D.A. 10.0.6 | Id: 1/1